

Kashyap Tanuku

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EDUCATION

Rutgers University

Bachelor of Science in Computer Science & Mathematics

New Brunswick, NJ

Sep 2022 – May 2026

Relevant Coursework

Machine Learning Principles, Artificial Intelligence, Data Science, Database Management, Data Structures & Algorithms, Design Analysis of Computer Algorithms, Combinatorics, Partial Differential Equations, Graph Theory, Linear Algebra, Computer Architecture.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, C#, R, HTML, CSS, SQL, Java

Frameworks/Tools: PyTorch, Kubernetes, Docker, Redis, vLLM, Neo4j, Faiss, Electron, TensorFlow, Linux, React, Angular, PyTest, FastAPI, WebSockets, Flask, Node.js, ChromaDB, MongoDB, AWS (EC2, S3, ECR, EKS, CloudWatch), Apache Spark, SQLite, Git, Ollama, scikit-learn, Hugging Face, .NET, Claude Code, Github Copilot, OpenCode, GPT Codex

EXPERIENCE

Research Assistant

Aresty Research Center

Aug 2025 – Present

New Brunswick, NJ

- Developed a cross-platform multiplayer experimentation system using **Unity** & **C#** to process concurrent non-blocking data streams, optimizing state synchronization latency to **under 500ms** across all participants.
- Engineered a real-time analytics pipeline using **C#** & **Firebase** to capture & log complex time-series data, enabling quantitative analysis on **500K+** experiment events.

AI/ML Engineer Intern

Coforge

Jun 2025 – Aug 2025

Princeton, NJ

- Built a RCA engine from **0 to 1** deployed via **Docker (ECR)** & **Kubernetes (EKS)** to automate failure diagnosis & remediation in large-scale microservices applications, **cutting SRE manual resolution time by 66%**.
- Created an ETL pipeline to process **1M+** telemetry data points with **PySpark** to train a custom GIN Model, using **PyG Graph Neural Networks** & Causal Inference, achieving **93% accuracy** in pinpointing service failures.
- Improved the quality of LLM-generated remediation plans by **117%** vs. naive RAG by engineering a custom RAG system with **LightRAG, Neo4j Knowledge Graphs, Redis, Faiss, & FastAPI**.
- Optimized LLM inference to deliver **20x higher throughput** vs. Hugging Face Inference & **under 2s P99** response latency using **vLLM** & **OpenAI API**.

Software Engineer Intern

BlaZop

Jun 2024 – Sep 2024

San Jose, CA

- Built an AI chatbot **end-to-end**, improving user-SaaS navigation for **5K+** users, with **Angular** & **Python**.
- Cut manual data collection efforts by **90%** by engineering custom **AI agents** to automatically generate **29K+** training instances from internal documentation using **Ollama**.
- Developed chatbot backend by fine-tuning a **LLaMA 3.1** model enhanced with RAG using **LangChain** & **ChromaDB** to provide context-aware responses based on enterprise knowledge bases.
- Created a service discovery search engine achieving **under 1s search latency** across **1K+** application architecture templates using **OpenAI CLIP** image embeddings & **ChromaDB**.

PROJECTS

Xpdite - Agentic AI Personal Assistant | Python, TypeScript, SQL, SQLite, Electron, FastAPI, MCP, LLMs, AI Agents

- Built a semantic MCP tool retriever using **vector embeddings** (Top-K selection) to dynamically route LLM queries to relevant tools at runtime (reducing context size), mirroring production tool-retrieval architectures used by frontier models like Claude.
- Engineered a multi-round agentic tool-calling loop (up to **50+ rounds**) across **6 MCP servers** (filesystem, terminal, websearch, Gmail, etc) via **FastAPI WebSockets**, enabling autonomous multi-step task execution across local & cloud environments.
- Designed a 4-layer LLM safety architecture for agentic terminal execution using blocklist enforcement, PATH injection protection, hard timeouts, & user approval prompts, mitigating prompt injection risks and unauthorized command execution.
- Implemented an Internal Memory System with **SQLite**-backed conversation persistence and token-aware context management, enabling context aware multi-session agentic behavior across local (Llama, Qwen) and cloud LLMs (**Claude, GPT, Gemini**).
- Architected an event-driven Cron Job Scheduler using **MCP** tool integrations to autonomously trigger remediation workflows in response to system events (crashes, errors), enabling self-healing agentic automation without human intervention.

Machine Learning Project - NBA Terminal | Python, Scikit-learn, NBA API, NumPy, Pandas

- Engineered an **XGBoost regressor** to predict NBA player point totals daily by analyzing **5K+** data points, achieving an **MAE of ~5** and incorporating custom features like Mean Reversion signal and a custom-made statistic: Zone Matchup Score.
- Developed a real-time analytics suite with **Python**, featuring a backtesting engine that validated a **56.2% win rate** on “Confident Bets” (predictions ± 3 points from season avg), outperforming the standard break-even threshold for sports betting.

Recommender Systems Project - Singular Value Decomposition | Python, NumPy, Pandas, Matplotlib, Seaborn, Apache Spark

- Engineered an SVD-based movie recommender using **Python** & **Apache Spark**, achieving **92% precision** / **96% NDCG** on the Netflix Prize Competition dataset.
- Developed a matrix factorization pipeline with **NumPy** & **Pandas**, predicting user ratings for unseen movies at **0.87 RMSE** / **0.67 MAE** & visualizing trends via **Matplotlib**.